
Tests and scans

Transrectal ultrasound scan (TRUS) and biopsy for prostate cancer

A transrectal ultrasound scan (TRUS) is an examination of the prostate gland using ultrasound. The doctor puts an ultrasound probe into your back passage (rectum).

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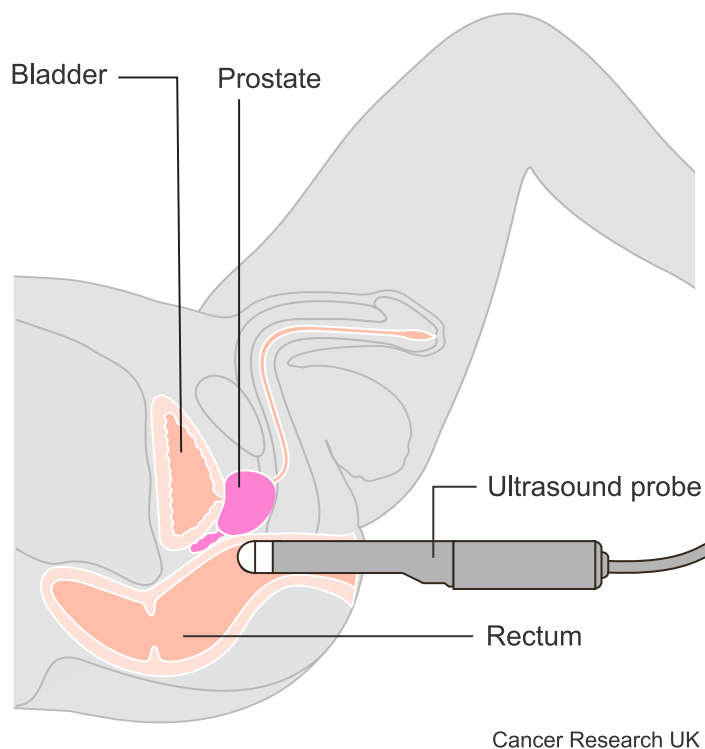
What is a TRUS?

Ultrasound scans use high frequency sound waves to create a picture of a part of the body. A prostate ultrasound scan can show changes in your prostate, including abnormal growths.

The ultrasound scanner has a microphone that gives off sound waves. The sound waves bounce off the organs inside your body, and the microphone picks them up. The microphone links to a computer. This turns the sound waves into a picture on the screen.



Trans rectal means that the doctor examines your prostate by putting the ultrasound probe into your back passage (rectum).



What is a TRUS guided biopsy?

A TRUS guided biopsy is when the doctor or nurse takes a sample of tissue during a TRUS. They use a thin needle to take the biopsy from the prostate through the wall of the rectum.

This type of needle biopsy looks for cancer cells in the prostate. Your doctor takes a series of small tissue samples from the prostate to examine under the microscope.

You have the biopsy through your back passage (rectum) using an ultrasound probe.

There are 2 main ways to take a biopsy from the prostate:

- TRUS guided biopsy
- transperineal biopsy - the needle goes through the perineum

Most hospitals in the UK now use transperineal biopsy to diagnose prostate cancer. TRUS biopsies are less common.

[Read about transperineal biopsy](#)

You usually have an MRI scan before your TRUS guided biopsy. The MRI scan helps your doctor decide whether you need a biopsy and where to take the biopsies from.

Preparing for your TRUS and guided biopsy

You usually have this test in the outpatient department using a local anaesthetic . Rarely, you may have it in the operating theatre under a general anaesthetic. This is when you are asleep and don't feel anything.

Your doctor or nurse will ask you to sign a consent form once they have given you information about the test.

You can't have a TRUS biopsy if you have a urine infection. Some hospitals might ask you to do a urine test to confirm.

You take antibiotics to stop an infection from developing after the biopsy. Your doctor will explain when you need to take the antibiotics and for how long. This is usually for a couple of days afterwards.

You need to make sure your bowel is empty when you go for your appointment. You might need to have an enema to empty your bowel. An enema is a liquid that you put into your back passage (rectum). Or you might have a liquid medicine to swallow the day before. You need to stay close to a toilet for a few hours after taking the medicine.

Check your appointment letter to find out how to prepare for your test.

Eating and drinking and medicines

You usually have a TRUS biopsy under local anaesthetic, so you can eat and drink normally beforehand and afterwards.

Take your usual medicines as normal, unless your doctor tells you otherwise. You need to stop taking drugs that thin your blood such as warfarin for a few days before your biopsy. Your doctor will tell you when you need to stop taking them. This is usually for 5 days before.

Tell your doctor if you also take drugs that stop cells in the blood called platelets from sticking together and forming a clot. These are antiplatelets. You need to stop taking these drugs for a few days before your biopsy.

What happens

Before

When you arrive at the department, a staff member asks you to take your lower clothes off and change into a hospital gown.

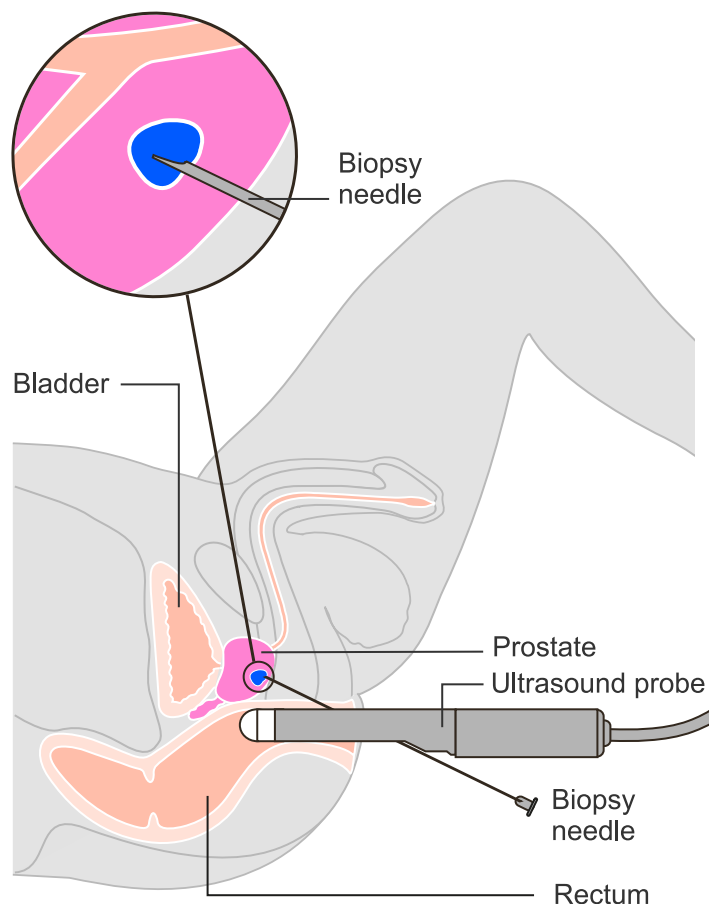
Before you have the TRUS, your doctor shows you the ultrasound machine. They also show you the very fine needle they use to take the tissue samples if you are having a TRUS guided biopsy. It can make quite a loud noise, so it's good to expect this.

Your doctor will explain what happens before they start, and you can ask any questions.

During the biopsy

You lie down on the scan table. They usually ask you to lie on your left side with your knees pulled up towards your chest. Your specialist injects local anaesthetic into your back passage (rectum), to numb the area.

The doctor puts an ultrasound probe into your rectum to examine your prostate. To get the samples of prostate tissue, they push the fine needle along the ultrasound probe and into the prostate gland.



Cancer Research UK

Your doctor uses the information from your MRI scan to guide where to take the biopsies from. Doctors call these targeted biopsies. They usually also take samples

from other areas of the prostate. These are systematic biopsies. If your MRI did not show any specific area that looks suspicious, then you only have systematic biopsies taken.

Your doctor will tell you how many biopsies they plan to take. The whole procedure takes about 20 to 30 minutes.

After

After the test, you have a rest and a drink. The team monitors you for at least 30 minutes afterwards. You can then go home. It's important to drink a lot of fluids for the next 24 hours.

Possible risks

TRUS and guided biopsies are safe tests but there are some possible risks. These might include:

- blood in your urine, poo and semen
- infection
- pain
- not getting enough biopsy samples

You usually have blood in your urine for several days afterwards. Drinking plenty of fluids helps to flush out the blood and helps to stop you from getting an infection.

Speak to our doctor or specialist nurse if you have signs of infection. This includes:

- a high temperature
- difficulty passing urine
- a need to pass urine very often
- a lot of blood in your urine or poo
- shivering or shaking

Getting your results

You won't get any results at the time of the scan. Any prostate tissue samples taken are sent to the laboratory to be looked at under the microscope. You should get your results within 1 or 3 weeks.

Waiting for test results can make you anxious. Ask your doctor or nurse how long it will take to get them. Contact them if you haven't heard anything after a couple of weeks.

You might have the contact details for a specialist nurse. You can contact them for information and support if you need to. It may help to talk to a close friend or relative about how you feel.



For support and information, you can call the Cancer Research UK information nurses. They can give advice about who can help you and what kind of support is available. Freephone: 0808 800 4040 – Monday to Friday, 9am to 5pm.

To talk in your preferred language, ask for our interpreting service.

References



- Transrectal ultrasound-guided biopsies of the prostate gland. Information about your procedure from The British Association of Urological Surgeons (BAUS)
The British Association of Urological Surgeons (BAUS), 2024
- Prostate cancer: diagnosis and management
National Institute for Health and Care Excellence (NICE), 2019. Last updated December 2021
- 2021 exceptional surveillance of prostate cancer: diagnosis and management (NICE guideline NG131)
National Institute for Health and Care Excellence (NICE), 2021
- Transrectal ultrasound biopsy of the prostate: does it still have a role in prostate cancer diagnosis?
A Moe and D Hayne
Translational Andrology and Urology, 2020. Vol 9, Issue 6. Pages 3018–3024

Last reviewed: 14 May 2025

Next review due: 14 May 2028

Related links

What is prostate cancer?



Prostate cancer is cancer of the prostate gland. The prostate gland is part of the male reproductive system, and is at the base of the bladder.

MRI and mpMRI scan for prostate cancer



A multiparametric MRI (mpMRI) is a special type of MRI scan. You usually have a mpMRI scan to help diagnose prostate cancer.

Tests for prostate cancer



You have a number of tests to check for prostate cancer. These may include a digital rectal examination, a PSA blood test, scans and a biopsy.

Transperineal biopsy for prostate cancer



Your doctor might take small samples of tissue from your prostate through the skin behind the testicles (perineum). This is a transperineal biopsy for prostate cancer.

Tests and scans



Find out about tests to diagnose cancer and monitor it during and after treatment, including what each test can show, how you have it and how to prepare.

Prostate cancer main page



Prostate cancer is cancer that starts in the prostate gland. The prostate gland is found at the base of the bladder and is about the size of a walnut.

Mental health and cancer

We know that it is common to struggle with your mental health when you have cancer or care for someone with cancer.

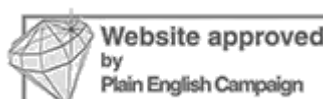
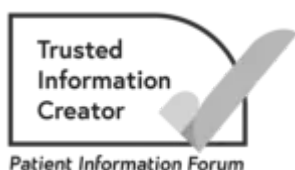
We have information to help



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